MILITARY STANDARD

TRUCKLOADING
WARHEAD, MK 103 MOD 0
(FOR MK 46 TORPEDO)
IN CONTAINER, MK 301 MODS 0 AND 1 AND
EXERCISE HEAD, MK 85 MODS 0, 1, 2, 3, 4, AND 6
DOMESTIC UNIT LOAD

MIL-STD-1320-132
(NAVY)

18 NOVEMBER 1988

SUPERSEDING

WR-51/132

18 DECEMBER 1970

UNIT LOAD DATA

DODIC/NALC:	T095 T618 T213 T088	T619	T091 T094	T092
LENGTH.....	48	48	48	48
WIDTH.....	41	41	41	41
HEIGHT.....	31-3/8	31-3/8	31-3/8	31-3/8
WEIGHT.....	815 LBS	830 LBS	794 LBS	885 LBS
CUBE.....	35.7 CU FT	35.7 CU FT	35.7 CU FT	35.7 CU FT

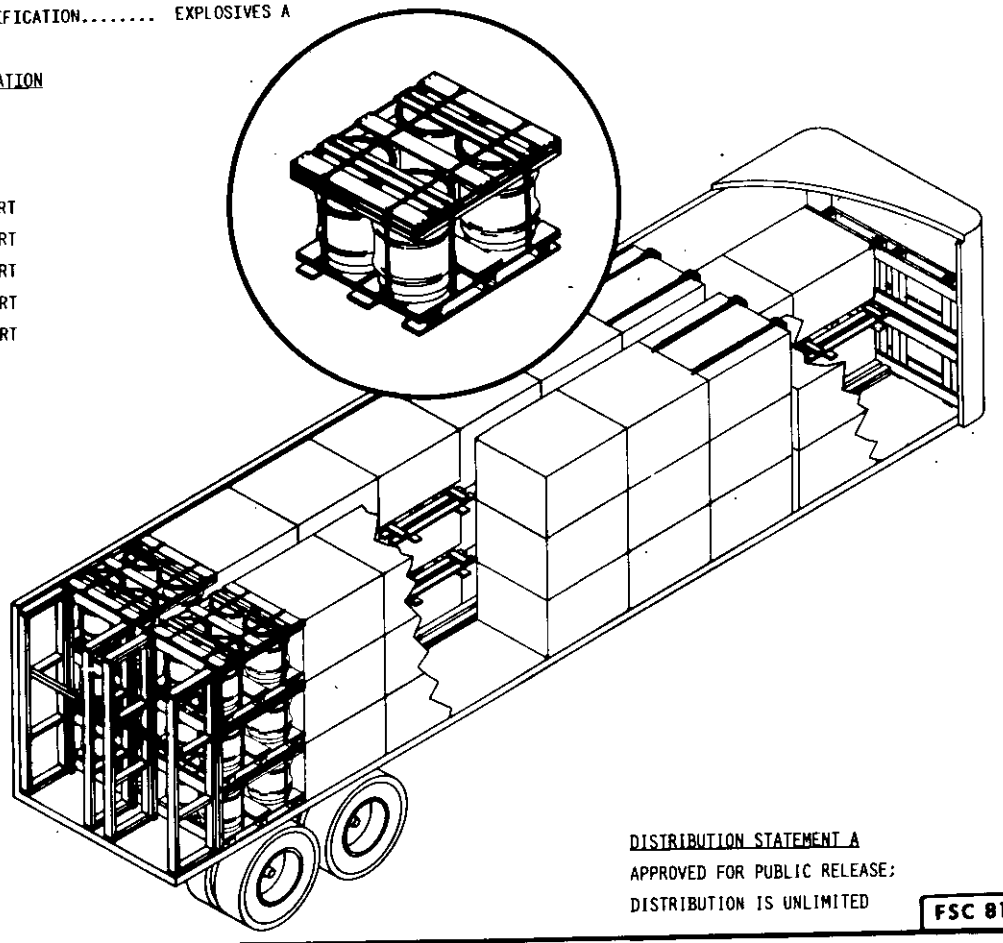
(MK 103 WARHEAD ONLY)

DOT HAZARD CLASSIFICATION..... EXPLOSIVES A

HAZARD CLASSIFICATION

DODIC/NALC	DOT
618	A
619	A
T088	INERT
T091	INERT
T092	INERT
T094	INERT
T212	INERT

AMSC NO. - N/A



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TECHNICAL DIRECTION AGENT (TDA) DATE

A.R. Quiros 11-18-88
BY DIRECTION DATE

ORIGINATOR William O Johnson 11/30/88
DATE

WPNSTA EARLE, NEW JERSEY

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GENERAL NOTES

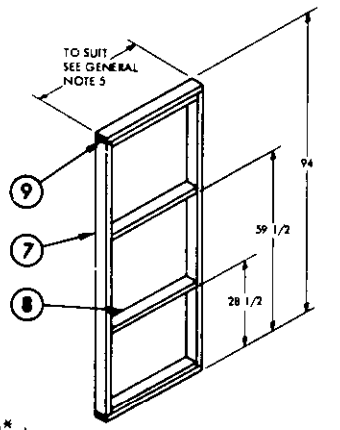
1. THIS DOCUMENT GIVES DETAILED INSTRUCTION IN TRUCKLOADING THE MK 103 MOD 0 TORPEDO WARHEAD AND MK 85 EXERCISE HEAD IN CONTAINER, MK 301 MOD 0 AND PALLETIZED PER MIL-STD-1322-792.
2. PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40 FT HIGH CUBE VAN-TYPE TRAILERS. DO NOT USE TRAILERS WITH METAL FLOORS.
3. SWAY BRACES FOR THE UPPER LAYERS OF UNIT LOADS SHALL BE INSTALLED AS TIGHT AS POSSIBLE. THE UNIT LOADS SHALL BE FIRMLY AGAINST THE SIDE WALLS OF THE TRAILER WITH THE SWAY BRACES HOLDING THE UNIT LOADS IN PLACE.
4. SWAY BRACE IS ASSEMBLED IN PLACE AS FOLLOWS:
   A. PREASSEMBLE PIECES NUMBER 1 AND 2 USING THREE 10 NAILS, CLINCHED, EACH JOINT. SPACER PIECE NUMBER 1 SHOULD BE LOCATED TO BEAR AGAINST PALLET STRINGERS WHEN IN POSITION.
   B. INSERT SUPPORTS, PIECE NUMBER 3, IN PLACE AND AGAINST INBOARD SIDE OF PALLET STRINGERS.
   C. SLIDE PREASSEMBLED PIECES NUMBER 1 AND 2 IN PLACE AND NAIL TO SUPPORTS, PIECE NUMBER 3, WITH ONE 6 NAIL EACH JOINT.
5. GATE FRAMES (DETAIL A) SHALL BEAR AGAINST REAR GATE AND TRAILER DOORS WHEN THE DOORS ARE CLOSED. SLACK SHALL NOT BE PERMITTED.
6. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
7. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO INSIDE OF TRAILER IN AN ACCESSIBLE AREA. CLOSE AND SEAL TRAILER DOORS AND ATTACH "EXPLOSIVES A" PLACARD TO BOTH SIDES, FRONT, AND BACK OF TRAILER.
8. APPLICABLE MATERIAL SPECIFICATIONS: DUNNAGE LUMBER, MM-L-751; NAILS FF-N-105, COMMON CEMENT COATED; STRAPPING QQ-S-781, TYPE 1, CLASS A.
9. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING WEAPONS REQUIREMENT MIL-STD-1320.

FTL 40 FT TRAILER (VAN TYPE)

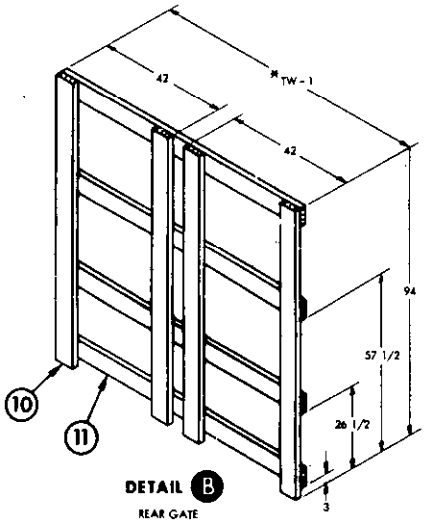
NOTE

ONLY A HIGH CUBE VAN OF A MINIMUM DOOR HEIGHT OPENING OF 98 INCHES WILL ACCOMMODATE THE THIRD LAYER OF THIS FTL PLAN. WHEN HIGH CUBE VANS ARE NOT AVAILABLE ONLY LESS-THAN-TRUCKLOAD QUANTITY CAN BE LOADED. SEE LTL INSTRUCTIONS.

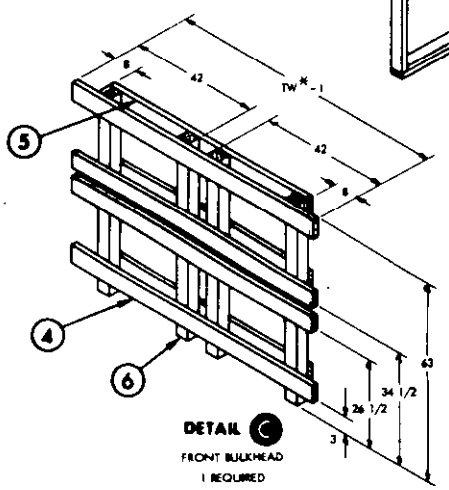
1. TO PREVENT LONGITUDINAL MOVEMENT IN THE THIRD LAYER, THE UNIT LOADS SHALL BE UNITIZED (SEE UNITIZING DETAIL, PAGE 7) WHERE THE LAYER CHANGES FROM THREE TO TWO LAYERS HIGH. THE STIFFENERS SHALL BE POSITIONED TOWARD THE DOUBLE LAYER SIDE.



DETAIL A
GATE FRAME
4 REQUIRED

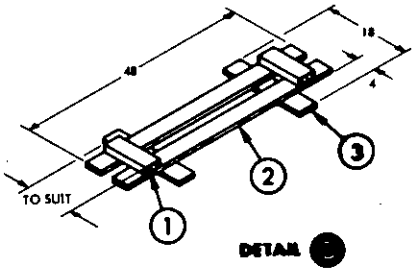


DETAIL B
REAR GATE
1 REQUIRED

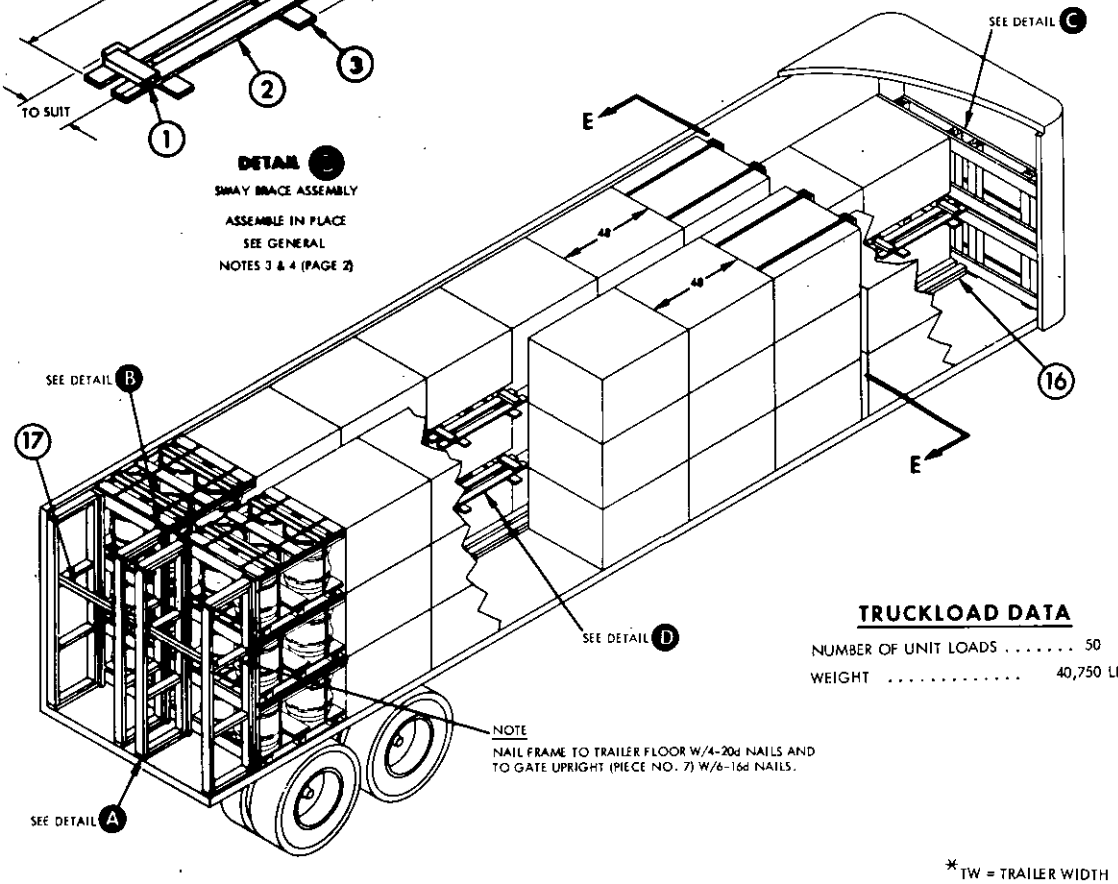


DETAIL C
FRONT BULKHEAD
1 REQUIRED

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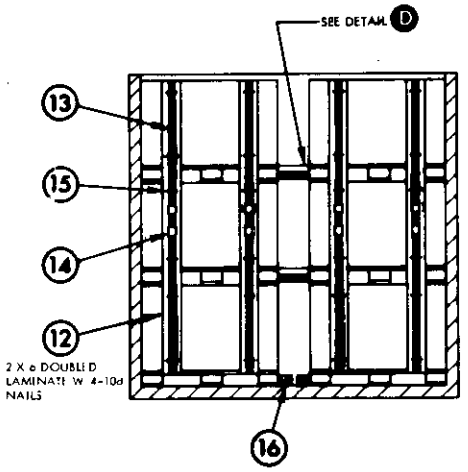
DETAIL A
SWAY BRACE ASSEMBLY
ASSEMBLE IN PLACE
SEE GENERAL
NOTES 3 & 4 (PAGE 2)



NOTE
NAIL FRAME TO TRAILER FLOOR W/4-20d NAILS AND TO GATE UPRIGHT (PIECE NO. 7) W/6-16d NAILS.

TRUCKLOAD DATA

NUMBER OF UNIT LOADS 50
WEIGHT . . . . . 40,750 LBS



2 X 4 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY EIGHT INCHES. NAIL 2ND PIECE TO 1ST PIECE IN LIKE MANNER.

SECTION E-E

*TW = TRAILER WIDTH

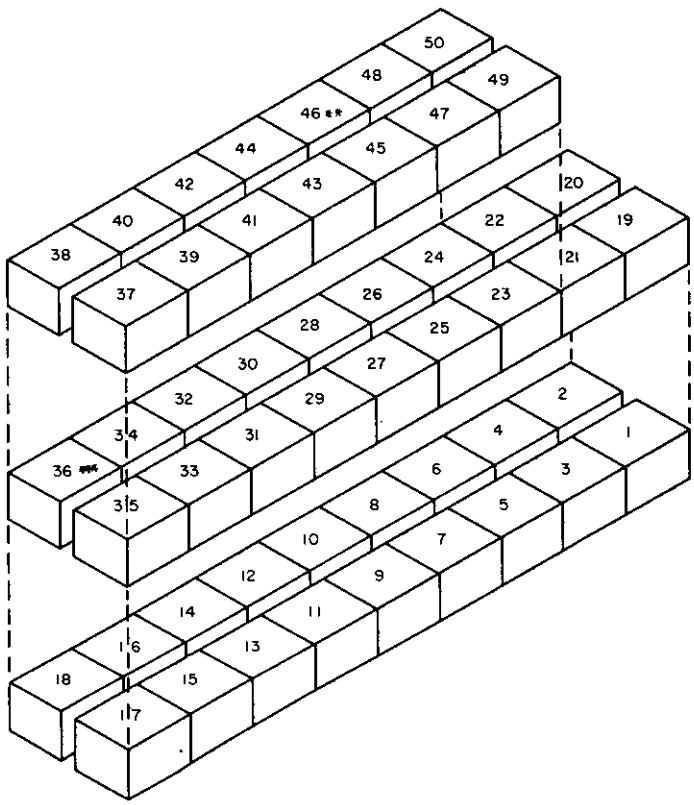
PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
	LIST OF MATERIALS AND NAILING DATA					
17	SPREADER	2 X 2 X TW*-1	1	7	2/JOINT	12d
16	SLEEPER	2 X 4 X 48	36	SEE SECTION E-E		
15	STAPLE	FOR 1 1/4 STRAP	24	-	-	-
14	SEAL	FOR 1 1/4 STRAP	8	-	-	-
13	STRAP	1 1/4 X .035 X 25 FT	4	-	-	-
12	STIFFENER	2 X 6 X 88 3/4	8	SEE SECTION E-E		
11	HORIZONTAL	2 X 6 X TW*-1	4	SEE 10	4/JOINT	10d
10	UPRIGHT	2 X 6 X 94	4	11	4/JOINT	10d
9	HORIZONTAL	2 X 4 TO SUIT	8	7	3/JOINT	12d
8	SPACER	2 X 4 TO SUIT	16	9	3/JOINT	8d
7	UPRIGHT	2 X 4 X 90 3/4	8	8	3/JOINT	12d
6	VERTICAL	4 X 4 X 63	4	SEE 4 & 5	-	-
5	FWD CROSS MEMBER	2 X 6 TO SUIT	4	6	3/JOINT	12d
4	AFT CROSS MEMBER	2 X 6 TW*-1	4	6	3/JOINT	12d
3	SUPPORT	1 X 4 X 18	32	SEE 2	-	-
2	STRINGER	1 X 4 X 48	32	3	1/JOINT	6d
1	SPACER	2 X 4 TO SUIT	32	2	3/JOINT	10d CLNHD

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LTL 40 FT TRAILER (VAN TYPE)

1. WHEN REQUIRED TO SHIP LESS-THAN-TRUCKLOAD, CONSULT THE LOAD PLAN BELOW. POSITION THE UNIT LOADS IN THE TRAILER FORMING THE SAME PATTERN AS THE NUMBER OF UNITS SHOWN IN THE PLAN (I.E., IF REQUIRED TO SHIP 30 UNIT LOADS, POSITION 30 LOADS AS INDICATED BY THE NUMBERS 1 THROUGH 30 IN THE PLAN. THE NUMBERS 31 AND ABOVE ARE NOT USED.)
2. LENGTH OF STRUT, PIECE NUMBER 8, SHALL BE SUFFICIENT TO CAUSE THE REAR BLOCKING ASSEMBLY TO BEAR AGAINST THE UNIT LOADS AND THE TRAILER DOORS WHEN THE DOORS ARE CLOSED.
3. WHEN REQUIRED TO SHIP 36 UNIT LOADS, THERE WILL BE TWO COMPLETE LAYERS. REAR BLOCKING SHOULD BE ACCOMPLISHED IN THE SAME MANNER AS THE FTL, PAGE 3, WITH THE REAR GATE ASSEMBLY REDUCED IN HEIGHT FOR A TWO LAYER LOAD.
4. THE LTL ON PAGE 5 IS FOR 26 UNIT LOADS. THE LTL ON PAGE 6 IS FOR 9 UNIT LOADS. BOTH SHOW THE CORRECT METHOD OF BLOCKING AND BRACING FOR THESE QUANTITIES.
5. ACTUAL SHIPPING REQUIREMENTS WILL DETERMINE THE NUMBER OF UNIT LOADS TO BE SHIPPED. THE LOAD PATTERN AND DUNNAGING SHALL BE IN ACCORDANCE WITH THE APPROPRIATE PARTS OF THE WR AND THE GENERAL DOCUMENT.

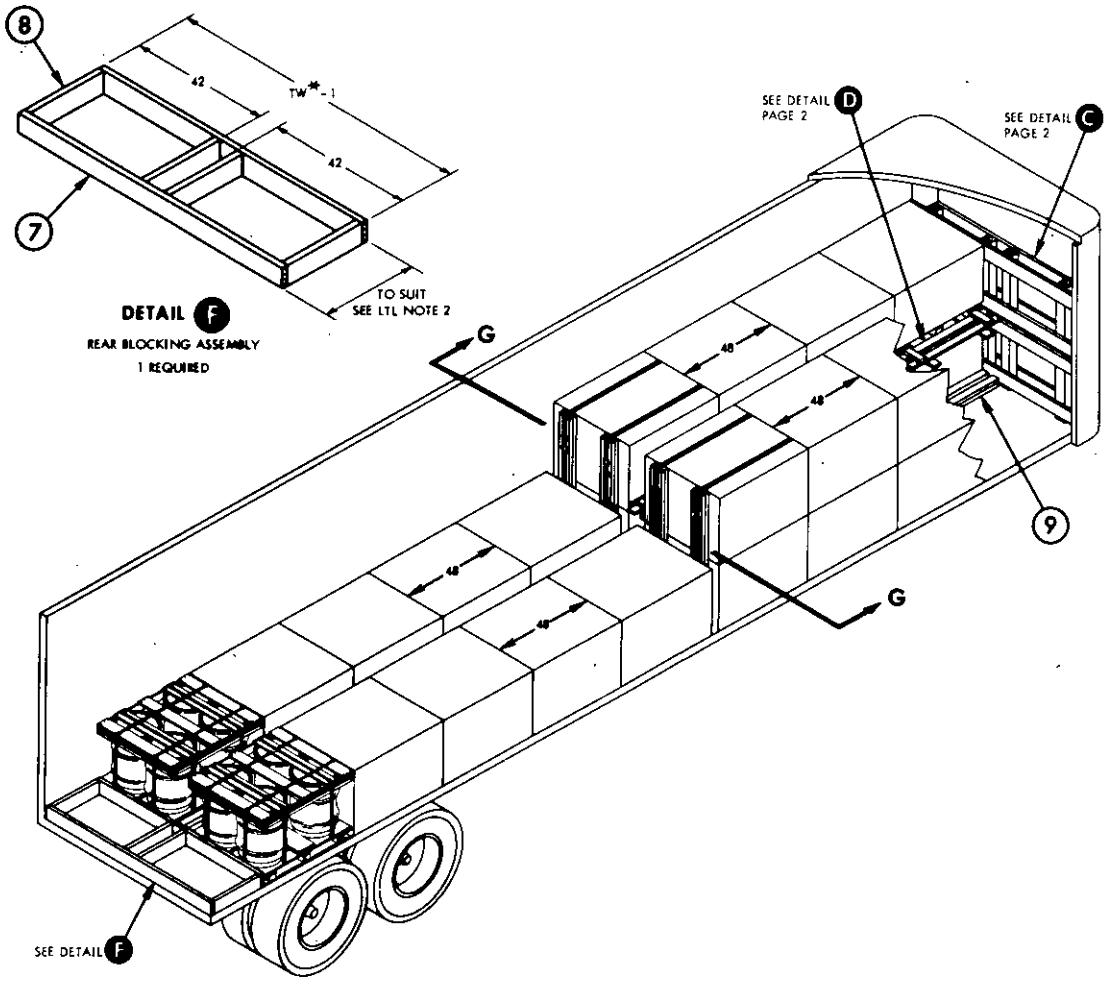
LOADING PLAN

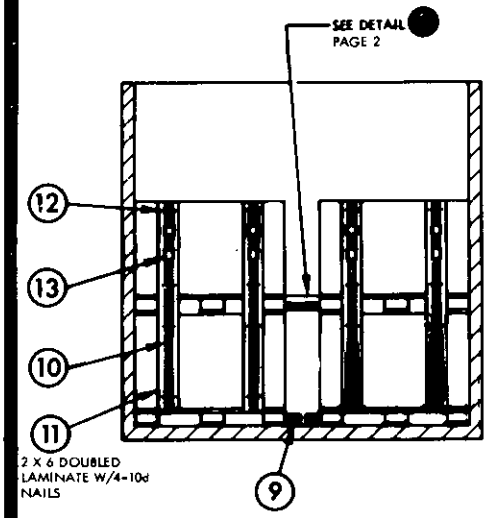


**WHEN EXERCISE HEAD NALC T092 OR MIXED LOADS OF VARIOUS NALCS ARE SHIPPED THAT INCLUDE NALC T092, FTL IS THEN CONSIDERED AS 46 UNIT LOADS.

***FTL FOR REGULAR CUBE VAN IS 36 UNIT LOADS.

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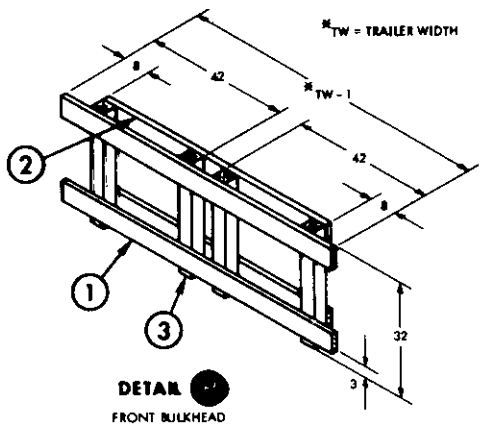


SECTION G-G

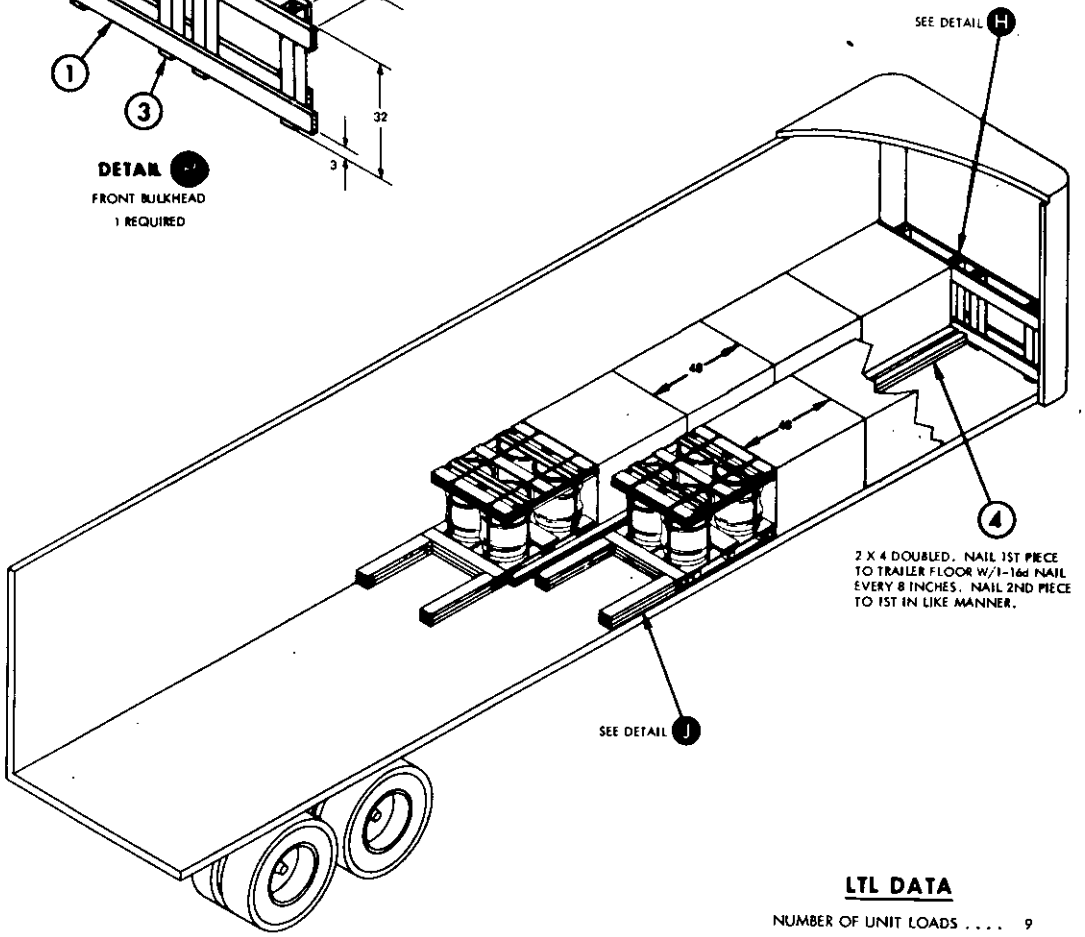
PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
13	SEAL	FOR 1 1/4 STRAP	8	-	-	-
12	STAPLE	FOR 1 1/4 STRAP	16	-	-	-
11	STIFFENER	2 X 6 X 59	8	SEE SECTION G-G		
10	STRAP	1 1/4 X .035 X 20 FT	4	-	-	-
9	SLEEPER	2 X 4 X 48	36	SEE SECTION G-G		
8	STRUT	2 X 6 TO SUIT	4	SEE 7	-	-
7	CROSS MEMBER	2 X 6 X TW*-1	2	8	3/JOINT	16d
6	VERTICAL	4 X 4 X 63	4	SEE 4 & 5	-	-
5	FWD CROSS MEMBER	2 X 6 TO SUIT	4	6	3/JOINT	12d
4	AFT CROSS MEMBER	2 X 6 X TW*-1	4	6	3/JOINT	12d
3	SUPPORT	1 X 4 X 18	8	SEE 2	-	-
2	STRINGER	1 X 4 X 48	8	3	1/JOINT	6d
1	SPACER	2 X 4 TO SUIT	8	2	3/JOINT	10d CLNHD

LIST OF MATERIALS AND NAILING DATA

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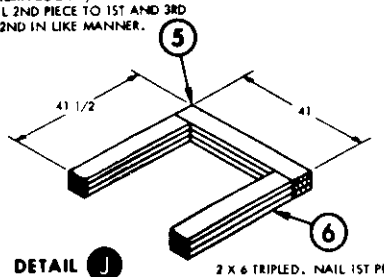
DETAIL H
FRONT BULKHEAD
1 REQUIRED



LTL DATA

NUMBER OF UNIT LOADS 9
WEIGHT ..... 7,335 LBS

2 X 6 TRIPLED. NAIL 1ST PIECE TO TRAILER FLOOR W/8-16d NAILS. NAIL 2ND PIECE TO 1ST AND 3RD TO 2ND IN LIKE MANNER.



DETAIL J
REAR BLOCKING
2 REQUIRED

2 X 6 TRIPLED. NAIL 1ST PIECE TO TRAILER FLOOR W/8-16d NAILS. NAIL 2ND PIECE TO 1ST AND 3RD TO 2ND IN LIKE MANNER.

*TW = TRAILER WIDTH

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
6	BACKUP CLEAT	2 X 6 X 36	12	SEE DETAIL J		
5	CROSS MEMBER	2 X 6 X 41	6	SEE DETAIL J		
4	SLEEPER	2 X 4 X 48	18	SEE FIELD NOTE		
3	VERTICAL	4 X 4 X 32	4	SEE 1 & 2	-	-
2	FWD CROSS MEMBER	2 X 6 TO SUIT	2	3	3/JOINT	12d
1	AFT CROSS MEMBER	2 X 6 X TW*-1	2	3	3/JOINT	12d

LIST OF MATERIALS AND NAILING DATA

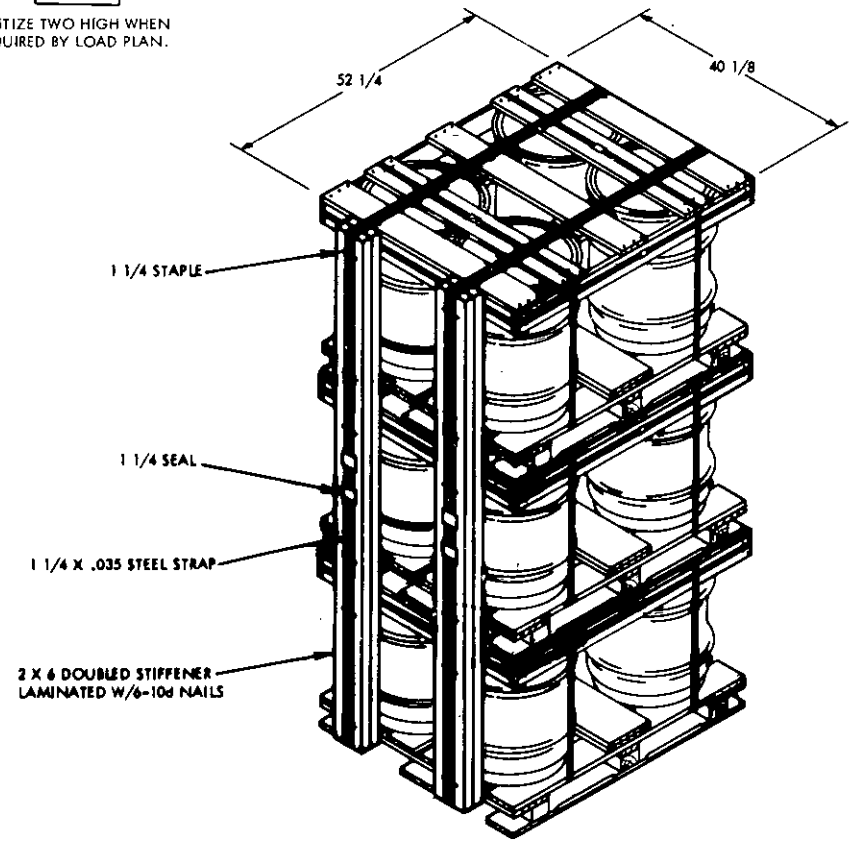
LTL 40 FT TRAILER (VAN TYPE)

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UNITIZING PROCEDURE

NOTE

UNITIZE TWO HIGH WHEN REQUIRED BY LOAD PLAN.



1. WHEN REQUIRED BY THE REQUIREMENTS OF THIS WR, THE TWO OR THREE HIGH STACKED UNIT LOADS SHALL BE STRAPPED AS SHOWN ABOVE.
2. THE 2 X 6 DOUBLED STIFFENER SHALL EXTEND TO THE TOP OF STACKED UNIT LOADS AND TO THE PALLET OF THE BOTTOM UNIT LOAD. THE STIFFENER SHALL BE LAMINATED WITH 10 NAILS TO PREVENT SHIFTING.
3. THE 1-1/4 INCH STEEL STRAPS POSITION AS SHOWN ENCIRCLE THE STACKED UNIT LOADS AND PASS UNDER THE DECK OF THE BOTTOM PALLET WITH THE STRAPS HOLDING THE STIFFENERS IN PLACE.
4. TENSION STRAPS AND SEAL WITH TWO DOUBLE CRIMPED SEALS. SECURE EACH STRAP TO THE STIFFENER WITH FOUR 1-1/4 INCH STAPLES.

REVIEW ACTIVITY:
NAVY - OS

PREPARING ACTIVITY:
NAVY - OS
(PROJECT NO. 8140-N801)

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